East Crazies EA Soils Resource Area Scope Of Work

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This scope of work is a Project SOW planning artifact for the East Crazy Inspiration Divide Land Exchange Environmental Assessment. It scopes soils-resource specialist work needed to support EA package preparation. It is not a final SOW award document, an applicability decision, a compliance finding, legal advice, a legal sufficiency determination, or a final agency decision.

Package Basis

Source artifacts reviewed:

- docs/PROJECT_SOW_PACKAGE_RUNBOOK.md
- docs/OUTPUT_SCHEMAS.md
- docs/PROJECT_SOW_OPERATIONALIZATION_ACCEPTANCE_MATRIX.md
- config/project_sow_resource_scopes_v1.json
- config/fixtures/project_sow/east_crazies_land_exchange_intake.json
- Generated East Crazies Project SOW package smoke output under /tmp/east-crazies-soils-sow-package/

Current package facts:

- Project: East Crazy Inspiration Divide Land Exchange Proposed Action
- Forest: Custer Gallatin National Forest
- District: Bozeman Ranger District
- NEPA level: environmental assessment
- Resource area: soil_resources
- Selected resource scope: vegetation_soils_air_quality
- Resource-scope discipline: vegetation_soils_air
- Soil-resource package status: covered by a selected SOW scope
- Current intake status: soil_resources is not explicitly listed as an expected proposed-action resource area
- Observed East Crazies calibration package status: no standalone soils specialist report is listed in observed_specialist_reports[]

Because the current intake covers soils through the combined vegetation, soils, air quality, climate, and carbon scope, this SOW starts with a soils screen. The specialist should decide whether a standalone soils effects report is needed or whether a concise no-issue or incorporated analysis memo is sufficient for the EA record.

Objective

Prepare a defensible soils-resource work product for the East Crazies EA that identifies whether the proposed land exchange, access commitments, trail work, wetland protections, grazing-related improvements, and parcel management changes could affect soil productivity, erosion, compaction, stability, revegetation, or sediment delivery. The work product must state the evidence basis, analysis boundary, mitigation or best-management-practice needs, and whether soils require a standalone specialist report.

Action Elements To Screen

The soils specialist should screen at least these proposed-action elements from the East Crazies intake:

- Federal and non-federal parcel exchange, including lands entering and leaving National Forest System ownership.
- Road and trail easements, Sweet Trunk Trail No. 274 construction, Inspiration Divide Trail No. 8 relocation, and Big Timber Canyon Trailhead improvement.
- Wetland, riparian, and monitoring-access deed restrictions.
- Grazing allotment, permit, fence, road, irrigation ditch, and related improvement changes.
- Hazardous-materials and site-condition due diligence before conveyance.
- Climate, carbon, vegetation, and restoration context where those topics depend on soil condition, productivity, or revegetation assumptions.

Required Tasks

1. Confirm the soils analysis boundary.
  - Identify federal disposal parcels, non-federal acquisition parcels, trail and trailhead work areas, access easements, improvement areas, wetlands/riparian buffers, and any construction or restoration footprints.
  - State whether soils analysis is limited to disturbance and restoration areas or also includes broader parcel management consequences after ownership transfer.

2. Compile baseline soils information.
  - Collect available soil survey, ecological site, slope, erosion-hazard, mass-movement, hydrologic soil group, road/trail, wetland/riparian, vegetation, and invasive-species data.
  - Identify fragile, highly erodible, compacted, saturated, shallow, unstable, or otherwise sensitive soil map units if present.
  - Record data gaps and whether field verification is required.

3. Screen proposed-action effects.
  - Evaluate soil disturbance from trail construction, trail relocation, trailhead improvement, road or trail easements, grazing-related improvements, and restoration commitments.

- Evaluate potential soil-productivity, compaction, displacement, erosion, sediment delivery, slope stability, reclamation, revegetation, and noxious-weed-prevention issues.
- Separate direct construction or implementation effects from long-term ownership and management changes.

4. Coordinate interdisciplinary dependencies.
  - Coordinate sediment and riparian questions with hydrology, wetlands, aquatics, and water quality.
  - Coordinate revegetation, restoration, carbon, invasive species, and whitebark pine or botany assumptions with vegetation and botany specialists.
  - Coordinate roads, trails, access, and recreation design assumptions with the recreation and engineering/access reviewers.
  - Coordinate parcel condition, hazardous-materials, mineral, and realty closing assumptions with lands/realty, minerals, and hazardous-materials reviewers.

5. Define mitigation, BMP, and monitoring requirements.
  - List erosion-control, sediment-control, soil salvage, decompaction, revegetation, weed prevention, seasonal timing, wet-weather operation, and monitoring measures needed for the EA and implementation record.
  - Identify which measures are design criteria, mitigation commitments, permit dependencies, or recommended implementation practices.

6. Prepare the soils record product.
  - If soils issues are present, prepare a soils effects technical memorandum or specialist report suitable for incorporation into the EA.
  - If soils issues are screened out, prepare a soils no-issue or incorporated-analysis memo with the evidence basis, assumptions, and responsible specialist signoff.
  - Include citations, maps or tables, unresolved data gaps, and an administrative-record source list.

Data Needs

- Parcel boundary, ownership, easement, and access commitment GIS layers.
- Proposed trail, trailhead, road, construction, relocation, and closure footprints.
- Wetland, riparian, stream, floodplain, and aquatic-resource overlays.
- Grazing allotment, fence, road, irrigation ditch, and improvement locations.
- Soil survey, ecological site, slope, erosion-hazard, mass-movement, and hydrologic soil data.
- Vegetation, invasive-species, revegetation, restoration, and carbon-analysis assumptions.
- Known design criteria, BMPs, permit conditions, monitoring commitments, and mitigation measures.
- Field notes, photo points, or site verification records if the specialist determines field review is needed.

Required Deliverables

- Soils resource screen and analysis-need determination.
- Soils data and assumptions table, including soil survey/map unit sources and disturbance-footprint assumptions.
- Soils effects memorandum or specialist report, or a documented no-issue/incorporated-analysis memo if a standalone report is not needed.
- Soil BMP, mitigation, restoration, and monitoring matrix.
- Interdisciplinary coordination notes for hydrology/wetlands, roads/trails/access, vegetation, realty, minerals, hazardous materials, and Forest Plan consistency dependencies.
- Administrative-record source list with citations and file locations.

Optional Deliverables

- Field verification plan, field forms, photo log, or map appendix.
- Erosion and sediment control checklist for trail, road, or trailhead implementation.
- Soil salvage, decompaction, revegetation, and monitoring plan outline.
- Soil sensitivity map exhibit for decision-record or implementation appendices.

Acceptance Criteria

- The product states whether soils require a standalone specialist report, incorporated analysis, or no-issue memo.
- Every screened soils concern is tied to a proposed-action element, parcel, or implementation footprint.
- Soil productivity, erosion, compaction, displacement, slope stability, sediment delivery, restoration, and invasive-species prevention are each either analyzed or explicitly screened out.
- Soil assumptions that support hydrology, wetlands, vegetation, carbon, roads/trails, realty, minerals, or hazardous-materials work are visible and coordinated.
- Mitigation, BMP, restoration, and monitoring commitments are listed as requirements, not assumed conclusions.
- The product includes enough citations, maps, tables, and administrative-record references for an EA reviewer to trace the analysis.

Assumptions

- Soil effects are not determined by the Project SOW package itself.
- The current East Crazies intake has no standalone observed soils specialist report and does not explicitly list soil_resources as a proposed-action-derived resource area.
- A soils specialist may determine that a standalone report is unnecessary if the record shows soil issues are adequately addressed through hydrology, wetlands, roads/trails/access, vegetation, carbon, hazardous-materials, or realty work products.

- Field verification needs remain subject to specialist judgment after reviewing the project footprint and available soil data.

Dependencies

- Reviewer-confirmed project footprint and implementation assumptions.
- Current parcel and access GIS data.
- Hydrology/wetlands and roads/trails/access design assumptions.
- Vegetation, restoration, carbon, and invasive-species assumptions.
- Realty, minerals, hazardous-materials, and closing-condition information.
- Forest Plan consistency direction for soil, watershed, restoration, or access-related standards and guidelines.

Reviewer Role And Timing

Reviewer role: soils specialist, or vegetation/soils/interdisciplinary resource specialist with documented soils competency.

Review timing:

- Initial soils screen before draft EA resource analysis is finalized.
- Update after trail, trailhead, access, wetland/riparian, and restoration footprints are confirmed.
- Final signoff before decision-record mitigation, design criteria, or implementation commitments rely on soils assumptions.

Recommended signoff fields:

- soils_analysis_boundary_confirmed
- soil_data_sources_confirmed
- field_review_need_confirmed
- soil_effects_pathway_confirmed
- bmp_mitigation_monitoring_confirmed
- standalone_soils_report_need_confirmed

Open Reviewer Decisions

- Should the East Crazies intake be amended to list soil_resources as an explicit proposed-action resource area?
- Is a standalone soils specialist report required, or can soils be documented through a no-issue memo or incorporated interdisciplinary analysis?
- Is field verification required for trail, trailhead, access, wetland/riparian, grazing improvement, or restoration areas?

- Which soil BMPs and monitoring commitments must be carried into the EA, decision record, and implementation file?